

AHA SCIENTIFIC STATEMENT

Advanced Molecular, Metabolic, and Imaging Approaches to Characterizing Right Ventricular Failure: A Scientific Statement From the American Heart Association

Soni Savai Pullamsetti, Chair, PhD, FAHA; Rebecca R. Vanderpool, PhD, FAHA; Frances de Man, PhD, FAHA; Vinicio A. de Jesus Perez, MD, FAHA; Anna R. Hemnes, MD; Monica Mukherjee, MD, MPH, FAHA; Laura Mercer-Rosa, MD, MSCE; Edda Spiekerkoetter, MD, FAHA; Khodr Tello, MD, PhD; Sebastien Bonnet, Vice Chair, PhD, FAHA; on behalf of the Council on Cardiopulmonary, Critical Care, Perioperative, and Resuscitation; Council on Clinical Cardiology; and Council on Cardiovascular Surgery and Anesthesia

ABSTRACT: Right ventricular (RV) dysfunction is a key predictor of outcomes in pulmonary hypertension (PH), substantially contributing to illness and death. As PH progresses, increased pulmonary vascular resistance places chronic pressure overload on the right ventricle. Initially, the right ventricle adapts through hypertrophic remodeling, thickening the heart wall to maintain cardiac output. Over time, this adaptive phase shifts to maladaptive remodeling, marked by RV dilation, fibrosis, stiffness, and decoupling from the pulmonary artery, known as RV–pulmonary arterial uncoupling. This uncoupling reflects the inability of the right ventricle to sustain contractility against elevated afterload, ultimately leading to right heart failure, the primary cause of death in late-stage PH. Awareness of RV dysfunction has grown, extending beyond PH and pulmonary arterial hypertension to systemic conditions, such as heart failure with preserved ejection fraction, congenital heart disease, COVID-19, and complications of left ventricular assist device implantation. Research is increasingly focused on understanding the molecular and hemodynamic drivers of RV failure, including inflammation and altered cellular signaling. Innovations in imaging and biomarker discovery are improving the detection of maladaptive RV remodeling. Promising treatments, such as the activin signaling inhibitor sotatercept, may reduce pulmonary vascular resistance and support RV recovery. Further work is needed to enhance RV function and prevent failure. This review summarizes current knowledge on RV dysfunction in PH, emphasizing its mechanisms, clinical relevance, and therapeutic potential. Recognizing the right ventricle as a central therapeutic target may lead to more personalized, effective interventions and improved patient outcomes in PH and related conditions.

Key Words: AHA Scientific Statements ■ biomarkers ■ heart failure ■ hypertension, pulmonary ■ omics/multiomics ■ ventricular remodeling

Right ventricular (RV) dysfunction is a key determinant of prognosis in pulmonary hypertension (PH), greatly affecting morbidity and mortality rates.¹ As PH progresses, the increased pulmonary vascular resistance (PVR) leads to chronic pressure overload on the right ventricle. Initially, the right ventricle compensates for this added burden through hypertrophic remodeling, a physiologic process where the right ventricle wall thickens to preserve cardiac output and maintain forward blood flow. This adaptation helps sustain function in the face of increasing resistance. However, prolonged

exposure to elevated PVR eventually results in maladaptive remodeling, characterized by RV dilation, fibrosis, RV diastolic stiffness, and RV–pulmonary arterial (PA) uncoupling.^{2,3} The uncoupling is driven by an inability of RV contractility to compensate for increased afterload. When this maladaptation occurs, it sets the stage for right heart failure (HF), which remains the leading cause of death in advanced stages of PH.

This review focuses primarily on RV failure arising from chronic pressure overload in PH, but it is important to recognize that RV dysfunction can also result

from acute insults, such as pulmonary embolism, acute respiratory distress syndrome, and sepsis. These acute conditions lead to rapid changes in RV loading conditions and can precipitate acute RV failure, which is beyond the scope of the current discussion but represents a clinical challenge.

Underappreciation of the vital role of the right ventricle in chronic HF has changed dramatically in recent years, highlighting the need for increased awareness and understanding.^{4–6} RV dysfunction is now recognized as a key factor not only in PA hypertension (PAH) but also in other systemic conditions, such as HF with preserved ejection fraction^{7,8}; congenital heart disease⁹; cardiomyopathies, such as dilated cardiomyopathy¹⁰; and acute infections, such as with SARS-CoV-2.¹¹ RV dysfunction is also a major cause of morbidity and mortality after left ventricular (LV) assist device implantation in patients with advanced LV dysfunction.^{12,13}

The recognition that RV dysfunction plays a pivotal role in the progression and severity of various cardiovascular diseases has led to more focused research into the molecular and hemodynamic mechanisms underlying RV failure. As a result, increasing emphasis has been placed on improving the molecular phenotyping of RV failure, with the goal of identifying the key signaling pathways and biomarkers that contribute to the onset and progression of RV failure. In addition, efforts are being made to improve the diagnosis of adaptive and maladaptive RV remodeling and RV failure through more precise imaging phenotyping. This is particularly important when the right ventricle is not the primary cause of RV failure, but temporary support of the right ventricle would be desirable until the primary cause can be fixed.¹⁴ The right ventricle demonstrates a notable ability to recover when afterload is reduced.¹⁵ With recent advances in the treatment of PAH—including drugs with potential antiremodeling properties, such as the activin signaling inhibitor sotatercept^{16–18}—there is optimism that PVR and RV afterload can be lowered substantially. However, until such therapeutic goals are fully realized, there is continued need for strategies aimed at enhancing RV function and preventing progression to RV failure.

This scientific statement provides a comprehensive synthesis of knowledge on RV failure specifically in the context of PH, focusing on the complex mechanisms underlying RV dysfunction, its clinical relevance, molecular drivers, advanced diagnostics, emerging management strategies, and potential therapeutic interventions. By synthesizing recent advances in the field, we aim to deepen understanding and inspire innovative approaches that can improve outcomes for patients with right ventricle–related complications in PH. Our goal is to establish the right ventricle as a novel therapeutic target, paving the way for more effective treatments for PH.

HEMODYNAMIC AND MOLECULAR MECHANISMS DRIVING RV DYSFUNCTION AND FAILURE

RV failure is a complex pathophysiologic process driven by a multifaceted interplay of hemodynamic stress, metabolic dysregulation, inflammation, fibrosis, and molecular signaling abnormalities (Figure 1). These mechanisms collectively disrupt RV contractility and efficiency, transitioning the heart from a compensatory to a decompensated state over time, greatly affecting patient outcomes.

Hemodynamic Stress and RV–PA Uncoupling

In PH, elevated PVR imposes continuous pressure overload on the right ventricle, leading to increased wall tension and initiating hypertrophic remodeling. However, RV afterload is multifactorial, and includes not only PVR but also pulmonary arterial compliance, characteristic impedance, and the pulsatile load exerted by the pulmonary arterial tree. These combined factors determine the total load against which the right ventricle must contract.

In many patients, early RV hypertrophy allows the ventricle to maintain cardiac output despite increased afterload, effectively compensating for elevated pressure. Over time, however, persistent hemodynamic stress may exceed the adaptive capacity of the right ventricle, resulting in RV–PA decoupling, where the right ventricle can no longer generate sufficient contractile force to match the resistive and pulsatile demands of the pulmonary circulation, leading to functional impairment.³ The ratio of RV end-systolic elastance (Ees) to arterial elastance (Ea) is a well-established measure of RV–PA coupling. Optimal coupling is generally observed at an Ees/Ea ratio between 1.5 and 2.0; an Ees/Ea ratio $< \approx 0.7$ indicates RV–PA uncoupling, a key physiologic threshold associated with RV failure.¹⁹ Clinically, uncoupling is associated with reduced stroke volume, elevated right atrial pressure, impaired perfusion, and increased susceptibility to ischemia–reperfusion injury and systemic venous congestion.

However, this model of linear progression from adaptive to maladaptive remodeling does not apply universally. Clinical observations reveal considerable heterogeneity: some patients display features of maladaptation early in the disease course and may not improve despite therapy, whereas others maintain RV adaptation for years, only to die of causes unrelated to PH. The mechanisms that govern whether an individual follows a progressive trajectory or displays an atypical pattern remain poorly understood. This variability underscores the importance of the focus of this scientific statement: to identify the determinants of RV adaptation and failure using advanced molecular, metabolic, and imaging approaches.

In more advanced or maladaptive stages of PH, RV dilation and functional tricuspid regurgitation can

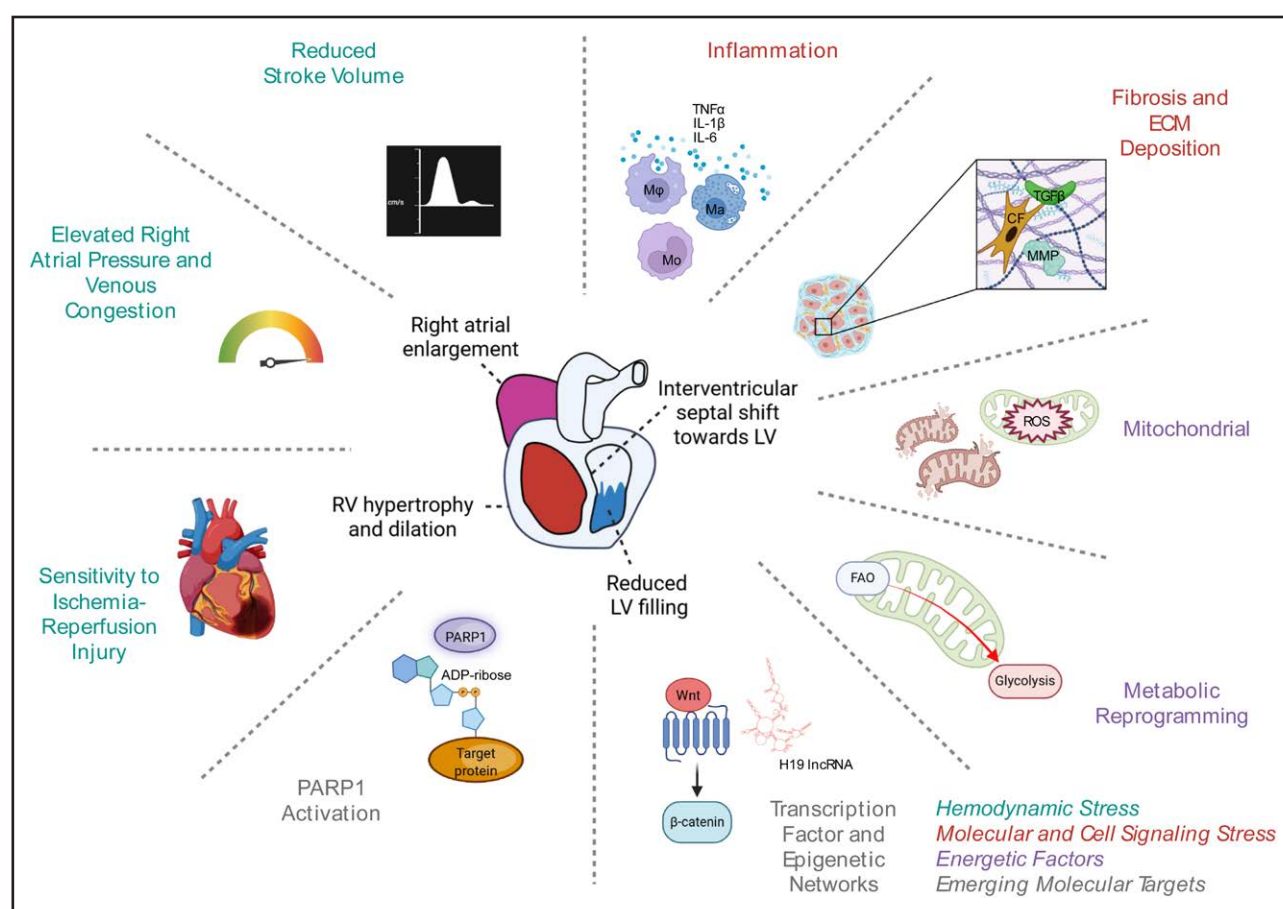


Figure 1. Mechanisms driving right ventricular dysfunction in pulmonary hypertension.

This schematic illustrates the multifactorial contributors to right ventricular (RV) dysfunction in pulmonary hypertension (PH), integrating anatomic and structural changes (central panel) with functional and molecular features of RV maladaptive remodeling, including hemodynamic stress, activation of stress signaling pathways, energetic dysregulation, and emerging therapeutic targets. Key mediators include inflammatory cytokines (eg, TNF- α [tumor necrosis factor- α], IL-1 β [interleukin-1 β], IL-6 [interleukin-6]), reactive oxygen species (ROS), transcription factors, epigenetic networks, and metabolic reprogramming (including dysregulated glycolysis and fatty acid oxidation). Molecular targets such as β -catenin, PARP1, Wnt signaling, and H19 long noncoding RNA are highlighted as central players in pathologic remodeling. These processes collectively drive mitochondrial dysfunction, fibrosis, extracellular matrix (ECM) deposition, DNA damage responses, and ultimately reduced stroke volume, RV dilation, venous congestion, and heightened sensitivity to ischemia–reperfusion injury. LV indicates left ventricle.

exacerbate hemodynamic instability and accelerate RV decompensation.

Inflammation and Fibrosis

Inflammation has increasingly emerged as a pivotal contributor to RV dysfunction in the context of PH. Elevated levels of inflammatory cytokines, such as TNF- α (tumor necrosis factor- α), IL-6 (interleukin-6), and IL-1 β (interleukin-1 β), are commonly observed in patients with PH and have been shown to correlate strongly with the severity of RV failure.^{20–23} In addition, recruitment of inflammatory cells—including macrophages^{24,25} and mast cells²⁶—is greatly increased during RV remodeling. Recent studies have identified monocyte-derived macrophages as key mediators of the proinflammatory environment in RV failure, in part through activation of the NLRP3 (NOD-, LRR-, and pyrin domain-containing protein 3) inflammasome.²⁷ These immune responses

contribute not only to sustained inflammation but also to maladaptive tissue remodeling.

Inflammation-driven activation of fibroblasts promotes pathologic extracellular matrix (ECM) deposition and RV fibrosis,^{28,29} which increases myocardial stiffness and impairs both systolic and diastolic function. This process establishes a self-perpetuating cycle of RV dysfunction. Key signaling pathways involved in this fibrotic response—including TGF- β (transforming growth factor- β), MMPs (matrix metalloproteinases), and integrins—have emerged as promising molecular targets.^{30–33} For example, upregulation of $\alpha 5 \beta 1$ integrin has been linked to increased RV fibrosis and contractile dysfunction in preclinical models of PH.³³

Translating these findings into clinical applications is an evolving area of research. Several anti-inflammatory and antifibrotic therapies are under investigation in preclinical and early-phase clinical trials. For instance, IL-1 (interleukin-1) and IL-6 blockade (eg, with agents

such as anakinra or tocilizumab) has shown efficacy in left HF and may hold potential for RV failure, particularly in patients with systemic inflammation. Likewise, emerging small molecule inhibitors and biologics targeting TGF- β signaling or integrin pathways could provide a novel therapeutic avenue for attenuating RV fibrosis in PH. Biomarkers derived from inflammatory or fibrotic signaling cascades may also support early identification of patients at risk for maladaptive remodeling and guide precision treatment approaches.

Ultimately, incorporating anti-inflammatory and anti-fibrotic strategies into the clinical management of PH could complement current vasodilator therapies and directly address RV pathology. Further clinical trials are needed to evaluate the safety, efficacy, and timing of such interventions, particularly in patients with early signs of RV maladaptation.

Mitochondrial Dysfunction and Metabolic Reprogramming

Mitochondrial dysfunction is another hallmark of RV failure in PH. Under normal physiologic conditions, the right ventricle primarily relies on fatty acid oxidation for energy production, which is an efficient pathway that meets the energetic demands of cardiac muscle.³⁴ However, during the compensatory hypertrophic phase, the right ventricle undergoes a metabolic shift toward glucose metabolism, a change that serves to support its increased energy demands during the initial stages of disease progression.^{35,36} This switch, although initially advantageous, becomes problematic as PH progresses, leading to a decline in mitochondrial efficiency. The right ventricle increasingly depends on glycolysis, even in the presence of sufficient oxygen, a phenomenon known as the Warburg effect. In addition, RV maladaptive remodeling leads to progressive accumulation of lipids in cardiomyocytes and lipotoxicity,³⁷ together with increased reactive oxygen species production in the myocardium,³⁸ further contributing to oxidative stress and ultimately impairing RV contractility. Mitochondrial dysfunction exacerbates the vulnerability of the right ventricle to ischemia–reperfusion injury,³⁹ a condition that accelerates the transition toward overt RV failure. As a result, therapeutic strategies focused on restoring fatty acid oxidation, enhancing mitochondrial biogenesis, or improving mitochondrial efficiency have shown promise in preclinical models.⁴⁰ These approaches represent a novel area for drug development aimed at tackling RV failure through metabolic modulation.

EMERGING MOLECULAR TARGETS IN RV FAILURE

Several distinct molecular pathways have surfaced as key contributors to RV dysfunction, offering potential

avenues for therapeutic intervention. PARP1 (poly[ADP-ribose] polymerase 1) overactivation is a noteworthy example: it promotes PKM2 (pyruvate kinase M2) activity, leading to a cascade of metabolic dysfunction, oxidative stress, and inflammation. Recent studies demonstrate that pharmacologic inhibition of PARP1 or modulation of PKM2 activity can confer protective effects against RV failure in rodent models subjected to pulmonary artery banding, highlighting a potential target for therapeutic intervention.⁴¹

Another promising target is the Wnt/ β -catenin signaling pathway, which has been implicated in the activation of cardiac fibroblasts and the promotion of RV fibrosis in the setting of PH. Inhibition of this pathway has shown potential to mitigate fibrosis and help preserve RV function.⁴² In addition, E2 (17 β -estradiol) or ER- α (estrogen receptor- α) agonist treatment attenuate PAH-induced RV failure by upregulating the procontractile and pro-survival peptide apelin through a BMPR2 (bone morphogenetic protein receptor 2)–dependent mechanism.⁴³ Neurohormonal activation through adrenergic receptors and modulation of the renin–angiotensin–aldosterone system are further approaches that have been tested in preclinical and clinical settings.⁴⁴

In addition to developing these new conceptual targets, researchers have also gained a better understanding of the role of disturbed BMP/TGF- β signaling, which plays a pivotal role in PAH. Patients with sequence variations in *BMPR2*, the most common sequence variation in hereditary PAH, are at increased risk of death and transplantation compared with noncarriers,⁴⁵ and have more severe RV dysfunction despite similar RV afterload.⁴⁶ Moreover, by combining RV phenotypes from cardiac magnetic resonance imaging (CMR) with genetic biomarkers in the UK Biobank, genetic variations in the *BMPR1A* locus were recently shown to be a female-specific genetic determinant of improved RV ejection fraction,⁴⁷ suggesting that the BMP/TGF- β pathway may serve as potential therapeutic target in different forms of PAH.

The immunosuppressive drug FK506 (tacrolimus) has been shown to reduce RV fibrosis in mouse models of chronic pressure overload, likely by a dual mode of action, involving both activation of BMP signaling and targeting of the immune system.⁴⁸ Inhibition of the TGF- β signaling pathway through an activin ligand trap is the most recent clinical development, with sotatercept being approved by the US Food and Drug Administration as a novel treatment for advanced PAH. In patients with PAH, sotatercept resulted in substantial improvements in PA pressures, PA compliance, PA–RV coupling, and RV function.^{17,49} In one study, sotatercept was shown to decrease RV mass and to increase hemoglobin and peak oxygen uptake, without changes in resting cardiac output.⁴⁹ This may be due to maintained systemic vascular resistance or to additional effects of sotatercept in other

organs, including the myocardium. Indeed, sotatercept was shown to ameliorate cardiac fibrosis and ejection fraction in a mouse model of LV chronic overload,⁵⁰ pointing to a general cardioprotective effect of sotatercept, which could be mediated directly on cardiomyocytes. The potential long-term consequences of this additional effect mediated by sotatercept on the right ventricle myocardium are unclear and deserve further attention.³ Identifying molecular pathways involved in RV adaptation or failure as key therapeutic targets marks a step forward to precision medicine for RV failure, offering new opportunities for treatments that directly address the underlying mechanisms of RV dysfunction.

DISTINGUISHING ADAPTIVE AND MALADAPTIVE RV REMODELING: INSIGHTS FROM IMAGING, HEMODYNAMICS, AND BIOMARKERS

Essential for determining a treatment window for RV dysfunction and failure is a clear definition and diagnosis of the transition from adaptive to maladaptive RV remodeling. In this section, we present in-depth phenotyping approaches to the right ventricle, focusing on hemodynamics, imaging, and biomarkers to distinguish and monitor this transition. We also highlight future directions for integrating molecular biomarkers with advanced imaging techniques, offering a more comprehensive understanding of disease pathogenesis and paving the way for personalized treatment strategies.

Definitions of Adaptive and Maladaptive RV Remodeling

In the context of PH, adaptive remodeling allows the right ventricle to cope with increased afterload primarily through cardiomyocyte hypertrophy. This adaptive process results in concentric RV hypertrophy, typically associated with minimal chamber dilation and limited fibrosis. Such adaptations enable the right ventricle to maintain systolic function and normal filling pressures, thereby enhancing contractility. This occurs by increasing the number of parallel sarcomeres, leading to greater force generation and enlarging the cross-sectional area of cardiomyocytes. This initial phase of adaptive remodeling is vital for sustaining cardiac output and preserving exercise capacity in response to heightened PA pressure. Although these adaptive changes are beneficial in the short term, they can lead to complications over time. Diastolic function may deteriorate due to increased stiffness stemming from interstitial fibrosis and altered mechanical properties of the right ventricle sarcomere, primarily driven by reduced β -adrenergic receptor-mediated phosphorylation of titin. Such changes can compromise the capacity of the right ventricle to fill adequately during

diastolic relaxation, setting off a detrimental cascade of effects.⁵¹

Maladaptive remodeling is characterized by substantial chamber dilation and insufficient contractility for a given afterload. This phase correlates with symptomatic HF, decreased exercise tolerance, elevated filling pressures, and progressive RV fibrosis. Patients experiencing maladaptive remodeling face a heightened risk of right HF, arrhythmias, and mortality.⁵² RV failure can compromise end-organ function by reducing oxygen delivery,⁵³ which has severe clinical consequences, highlighting the importance of early intervention in therapies targeting RV maladaptive remodeling.

The mechanism behind the transition from adaptive to maladaptive remodeling remains unclear, but is influenced by various disease pathogeneses, including congenital heart disease, left heart disease, and chronic lung disease, as well as the severity of afterload experienced by the right ventricle. In addition, patient-specific factors, such as age, sex, early life exposures, lifestyle factors, comorbidities, and genetic predisposition, also play vital roles in modulating the capacity of the right ventricle to respond to hemodynamic stress, and influence this transition. The transition point remains elusive due to delayed diagnosis when patients already exhibit maladaptive RV function and limited longitudinal patient data. Recent advancements in imaging technologies, particularly CMR, alongside important work from the Hopkins group^{53a, 53b} on RV contractile reserve have revealed that the transition is more closely linked to the ability of the right ventricle to enhance contractility than to the degree of afterload alone.⁵⁴ Therefore, quantifying RV-PA coupling, which provides valuable insights into the adaptive capacity of the right ventricle in response to increasing pressures, is key for assessing symptoms and prognostic outcomes in patients with PH.

Imaging and Hemodynamic Measures to Differentiate Between Adaptive and Maladaptive RV Remodeling

Various echocardiographic and hemodynamic measures are essential for differentiating between adaptive and maladaptive RV remodeling. The gold standard for assessing RV-PA coupling is the Ees/Ea ratio, typically measured using RV conductance catheters. Optimal coupling is generally observed at an Ees/Ea ratio between 1.5 and 2.0; an Ees/Ea ratio ≤ 0.7 indicates RV-PA uncoupling, a key physiologic threshold associated with RV failure.¹⁹ Given the invasive nature of pressure-volume loop analysis, there is growing interest in developing noninvasive imaging techniques to derive surrogate markers of RV-PA coupling.

Echocardiographic metrics, such as the ratio of tricuspid annular plane systolic excursion (TAPSE) to PA systolic pressure, serve as key noninvasive indices of RV-PA

coupling, providing prognostic value across various clinical scenarios, including precapillary and postcapillary PH and HF.⁵⁵ However, TAPSE predominantly reflects longitudinal shortening, and can be influenced by multiple factors, such as angulation, preload, and the overall loading conditions of the heart, which limit its effectiveness in accurately distinguishing between remodeling types. TAPSE often decreases during early remodeling, but may remain unchanged in advanced disease stages,¹⁹ further complicating the assessment of RV function.⁵⁶

Emerging echocardiographic measures, including RV free wall strain to PA systolic pressure ratio, demonstrate promise in overcoming the limitations inherent to TAPSE measurements. Recent studies suggest that RV free wall strain may be superior to TAPSE/PA systolic pressure in predicting clinical outcomes in patients with PAH.⁵⁷

The complex shape and contraction dynamics of the right ventricle make conventional assessment techniques, particularly those derived from 2-dimensional imaging, challenging. Three-dimensional volumetry obtained through echocardiography or CMR allows for a more comprehensive evaluation, revealing dynamic changes in contraction patterns that may indicate remodeling before substantial declines in ejection fraction or increases in cardiac volume occur. Among these, RV ejection fraction, particularly changes in RV ejection fraction in response to therapy, has gained recognition as a robust marker of disease progression and treatment efficacy,⁵⁵ as highlighted in the work of Von Noordegraaf and colleagues.^{57a} Considering changing population demographics and the higher expected prevalence of obesity in the coming years, CMR will be a good alternative approach for obtaining insights into RV adaptation.⁵⁸

Beyond volumetric and functional measures, CMR also enables tissue characterization. Techniques such as late gadolinium enhancement and T1/T2 mapping allow for the detection of myocardial fibrosis and inflammation, both of which are associated with maladaptive RV remodeling and worse clinical outcomes. Four-dimensional flow imaging has the potential to further clarify changes in RV and right atrial morphology over time and may also provide a reliable prediction of PH when compared with the gold standard right heart catheterization.⁵⁹ Four-dimensional flow imaging also could be used to assess the impact of tricuspid insufficiency on right atrial function and identify congenital anomalies that may underlie or complicate RV dysfunction.

Cardiac positron emission tomography (PET) imaging provides further mechanistic insights by identifying metabolic shifts in the right ventricle. For example, a transition from oxidative phosphorylation to glycolysis—a hallmark of metabolic remodeling—can be detected through PET and has been linked to maladaptive phenotypes in PH. This metabolic signature may serve as an early marker of RV dysfunction and a potential target for emerging therapies. In addition, tracers such as ⁶⁸Ga-Dotatate

(which reflects inflammation) and ¹¹C-CGP-12177 (which assesses β -adrenergic receptor expression) could be used to validate and translate pathophysiologic mechanisms.

As the clinical burden of obesity and comorbidities increases, modalities such as CMR and PET, which offer high-resolution, reproducible assessments independent of body habitus, will play an increasingly important role in evaluating RV adaptation and guiding patient management.⁵⁸

Metabolic and Molecular Targets to Differentiate Between Adaptive and Maladaptive RV Remodeling

Metabolic Biomarkers

Metabolic adaptations of the adult right ventricle differ between adaptive and maladaptive remodeling in the context of PH (Figure 2). In a healthy adult heart, fatty acids serve as the primary substrate for mitochondrial oxidation, efficiently supporting energy production. However, during fetal development, when circulating fatty acids are low, glycolysis and glucose oxidation dominate adenosine triphosphate production. As adults experience increased afterload, complex metabolic shifts occur, often studied in animal models, particularly during the early stages of RV hypertrophy. During the initial phases of adaptive remodeling, the right ventricle partially recapitulates fetal gene expression patterns, leading to metabolic changes that favor glucose utilization over fatty acid metabolism. Research using rodent models has shown that this early adaptive response is characterized by enhanced glucose uptake and increased glycolytic activity. These adaptations are crucial for sustaining adenosine triphosphate production despite reduced oxygen delivery, which becomes vital during the early stages of heightened afterload.^{34,60}

These metabolic shifts can be beneficial in the short term, but may lead to increased production of mitochondrial reactive oxygen species, contributing to oxidative stress and the gradual progression of RV failure over time. As RV dysfunction progresses, a reduction in fatty acid oxidation is often observed, resulting in cardiomyocyte lipid deposition and the accumulation of lipotoxic ceramides. This metabolic transition exacerbates the decline in cardiac function and can lead to further adverse remodeling. Recent studies have reported a notable decrease in tricarboxylic acid cycle gene expression during RV decompensation in human samples, highlighting the considerable metabolic derangements that arise in advanced stages of the disease.⁶¹ Advanced imaging techniques have similarly revealed metabolic changes, including enhanced glucose uptake and lipid accumulation in the right ventricle of patients with PH.^{37,60}

Circulating tricarboxylic acid metabolites, such as succinate and lactate, have been linked to impaired mitochondrial function in the right ventricle.⁶² Elevated levels

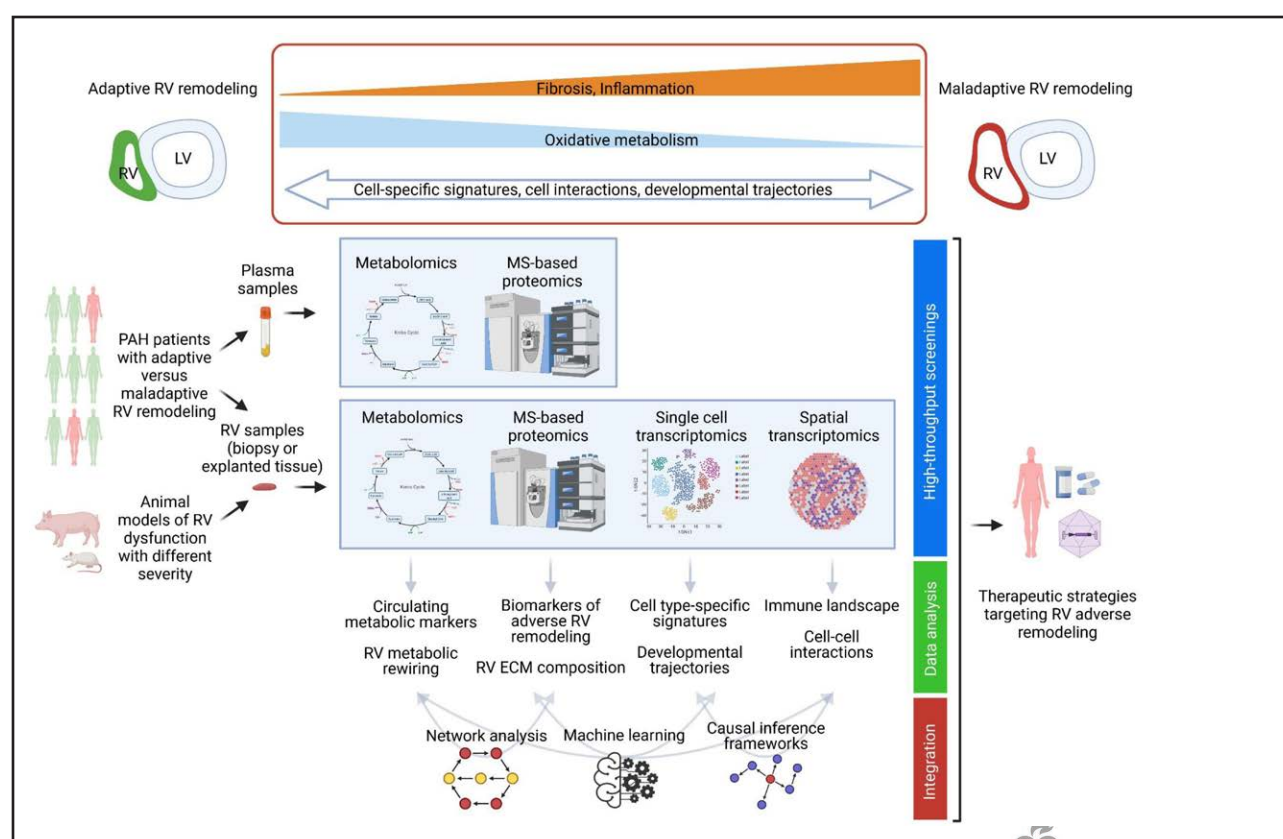


Figure 2. Multi-omics and systems biology approaches to decipher right ventricular remodeling in pulmonary arterial hypertension.

This schematic outlines an integrative strategy to dissect adaptive versus maladaptive right ventricular (RV) remodeling in pulmonary arterial hypertension (PAH) using samples from patients (plasma, biopsies, or explanted RV tissue) and animal models of RV dysfunction of varying severity. The approach combines mass spectrometry (MS)–based proteomics, metabolomics, and spatial transcriptomics to define cell type–specific signatures, immune landscape, extracellular matrix (ECM) composition, metabolic rewiring (eg, Krebs cycle alterations), and developmental trajectories. The double arrow indicates that the potential changes induced by maladaptive remodeling are still under active investigation. These data help identify biomarkers of adverse remodeling, circulating metabolic markers, fibrosis, inflammation, and oxidative metabolism changes. Data integration is achieved through systems biology approaches aimed at identifying regulatory nodes or pathways causally linked to disease pathoprogession. This systems-level framework supports the development of therapeutic strategies targeting maladaptive RV remodeling. LV indicates left ventricle.

of succinate, for example, are associated with increased oxidative stress and inflammation, both of which contribute to the progression of RV dysfunction. Succinate accumulation can serve as an indicator of mitochondrial dysfunction, reflecting the underlying pathophysiology related to energy deficits. Similarly, lactate accumulation indicates a reliance on glycolysis even in the presence of oxygen, exacerbating energy deficits in the right ventricle and highlighting a shift away from efficient adenosine triphosphate production.⁶³ Monitoring these metabolic biomarkers offers insights into the metabolic health of the right ventricle and may aid in the identification of patients at higher risk for rapid disease progression. Such an approach may facilitate early intervention strategies aimed at restoring normal metabolic functions.

Given the importance of understanding the transition from RV adaptation to dysfunction, researchers are actively exploring plasma metabolites as potential biomarkers for RV function and early indicators of dysfunction.^{63,64} Promising molecular mediators of RV dysfunction

are presented in Table 1. Emerging evidence suggests that metabolites in the kynurenine pathway may correlate with RV dysfunction in PAH. In addition, elevated plasma acylcarnitines, which serve as markers of fatty acid transport, have been associated with increased lipid content in the right ventricle among patients with PAH.⁶⁵ Despite these promising findings, the relationship between these metabolic changes and the transition from adaptive to maladaptive remodeling remains to be fully elucidated. Ongoing research is essential to determine how these metabolites can serve as reliable markers for monitoring RV function and identifying early signs of dysfunction in patients with PH, thereby facilitating timely intervention and effective management strategies.

Long Noncoding RNAs as Biomarkers for RV Adverse Remodeling

One of the most promising biomarkers under investigation is H19, a long noncoding RNA that is crucial in cellular stress responses and tissue remodeling. H19

Table 1. Promising Molecular Mediators and Biomarker Candidates of Right Ventricular Dysfunction

Pathways or nodes	RV-relevant takeaways	Evidence context	References
Signaling pathways (TGF- β /activin/BMP, Wnt, NO, hormones, inflammasome)			
TNF- α /IL-6/IL-1 β (inflammatory cytokines)	Systemic and myocardial cytokines track RV dysfunction; myeloid IL-1 β contributes to PH (implicates RV load and injury)	Human heart transplant, PAH cohorts; HFpEF-PH mouse	20–23
NLRP3 inflammasome (macrophage)	Macrophage NLRP3 activation promotes RV failure in PAH; therapeutic axis	Human PAH RV tissue and rodent models	27
eNOS/NO–collagen crosstalk	eNOS signaling tied to RV remodeling and collagen metabolism in PH	Rat models of PH	24
TGF- β superfamily (TGF- β /activin vs BMP)	Maladaptive fibrosis and remodeling through TGF- β ; precision approaches targeting soluble TGF- β /activin ligands show promise	Reviews and translational updates; PAH context	30, 31
BMP/BMPR2 restoration	BMPR2 signaling preserves right ventricle; boosting BMP (eg, FK506) improves RV function; estradiol–ER α benefits right ventricle through BMPR2/apelin; BMPR2 genotype affects outcomes and RV phenotype	Human genetics, clinical cohorts, rodent and cell studies	43–47
Activin trap (sotatercept class)	Improves exercise tolerance and RV function in PAH; analog benefits cardiopulmonary remodeling in LHF-PH models	Phase 2b PAH study; preclinical LHF-PH	49, 50
Wnt– β -catenin–FOSL	Targeted inhibition ameliorates RV remodeling through fibrotic and metabolic programs	Preclinical RV overload models	42
PARP1–PKM2 axis (immunometabolic)	PARP1 controls PKM2 nuclear shuttling, driving profibrotic and proinflammatory gene expression in right ventricle; inhibition restores metabolic homeostasis and function	Human RV tissue; monocrotaline/SU5416 models	34–38
Sex hormones (estradiol, ER α , and androgens)	Estradiol–ER α signaling enhances RV adaptation (through apelin, BMPR2 stabilization); androgen pathways may worsen maladaptive remodeling	Human cohorts with sex-stratified outcomes; preclinical hormone manipulation	44–46
Hippo–YAP/TAZ	YAP/TAZ activity regulates cardiomyocyte proliferation, fibroblast activation, and capillary density; overactivation linked to maladaptive remodeling	Preclinical RV pressure-overload models	32, 33
PI3K–Akt–mTOR	Supports early compensatory hypertrophy and angiogenesis; chronic overactivation contributes to maladaptation	Rodent pressure overload; PAH RV tissue	30, 41
ECM and fibrosis pathways			
Collagen I/III, crosslinking enzymes (LOX, PLOD2)	Increased collagen deposition and crosslinking stiffen the right ventricle and impair diastolic function	Human PAH RV biopsies; SU5416-hypoxia rat	11–13
MMPs (MMP-2, MMP-9)	Dysregulated ECM turnover promotes maladaptive remodeling; MMP inhibition partly protective	Rodent PH models; human plasma biomarker studies	15, 16
TIMP–MMP balance	TIMP overexpression skews toward fibrosis and reduced compliance	PAH myocardium; experimental PH	16, 17
Integrin–FAK signaling	Links mechanical stress to fibroblast activation and ECM deposition; integrin blockade reduces RV fibrosis	RV pressure overload mouse models	18, 19
Periostin	Upregulated in maladaptive RV; promotes fibroblast activation and ECM stiffening	Human failing ventricles, rodent PH	28
Galectin-3	Circulating galectin-3 tracks RV dysfunction; drives fibroinflammatory responses	PAH and HF cohorts; preclinical inhibition studies	25, 26
Metabolic or mitochondrial pathways			
PGC-1 α /PPAR α axis	PGC-1 α downregulation reduces mitochondrial biogenesis; PPAR α agonists restore FAO and improve RV function	Human failing RV tissue; MCT and hypoxia models	37, 38
FAO–glycolysis switch	Right ventricle shifts toward glycolysis with impaired FAO; mimics Warburg-like state	Rodent PH models; human RV biopsies	38, 39
PDK4–PDH inhibition	PDK4 upregulation inhibits PDH, worsening glycolytic reliance; dichloroacetate improves RV energetics and contractility	Rodent PH and RV pressure overload; pilot human studies	39, 40
HIF-1 α stabilization	Drives glycolytic reprogramming and angiogenic mismatch; chronic HIF activation maladaptive	Preclinical PH; human RV samples	41
Mitochondrial dynamics (DRP1, OPA1)	Excessive fission (DRP1) contributes to RV dysfunction; inhibition of DRP1 preserves contractility	Rodent RV overload models	42
Iron–heme metabolism	RV iron handling defects contribute to energetic stress; iron repletion shows benefit in PAH	PAH clinical trials; RV tissue analysis	29

(Continued)

Table 1. Continued

Pathways or nodes	RV-relevant takeaways	Evidence context	References
Ketone metabolism	Ketone oxidation upregulated in failing right ventricle; may act as adaptive fuel reserve	Human PAH RV samples; rodent PH	³⁸

Akt indicates protein kinase B; BMP, bone morphogenic protein; BMPR2, bone morphogenic protein receptor 2; DRP1, dynamin-related protein 1; ECM, extra-cellular matrix; eNOS, endothelial nitric oxide synthase; ER α , estrogen receptor α ; FAK, focal adhesion kinase; FAO, fatty acid oxidation; FOSL, FOS-like antigens; HFpEF-PH, pulmonary hypertension in heart failure with preserved ejection fraction; HIF-1 α , hypoxia-inducible factor-1 α ; IL-1 β , interleukin-1 β ; IL-6, interleukin-6; LHF-PH, pulmonary hypertension secondary to left heart failure; LOX, lysyl oxidase; MCT, monocarboxylate transporter; MMP, matrix metalloproteinase; mTOR, mam-malian target of rapamycin; NLRP3, NOD-, LRR-, and pyrin domain-containing protein 3; NO, nitric oxide; OPA1, optic atrophy 1; PAH, pulmonary arterial hypertension; PARP1, poly(ADP-ribose) polymerase 1; PDH, pyruvate dehydrogenase; PDK4, pyruvate dehydrogenase kinase 4; PGC-1 α , peroxisome proliferator-activated receptor γ coactivator-1 α ; PH, pulmonary hypertension; PI3K, phosphatidylinositol-3 kinase; PKM2, pyruvate kinase M2; PLOD2, procollagen-lysine, 2-oxoglutarate 5-dioxygenase 2; PPAR α , peroxisome proliferator-activated receptor α ; RV, right ventricular; TGF- β , transforming growth factor- β ; TIMP, tissue inhibitor of metalloproteinases; TNF- α , tumor necrosis factor- α ; and YAP/TAZ, Yes-associated protein/transcriptional coactivator with PDZ-binding motif.

has been found to be upregulated in patients exhibit-ing decompensated RV function, particularly those with maladaptive remodeling. H19 does not translate into proteins, but exerts regulatory functions at both tran-scriptional and posttranscriptional levels, influencing key molecular pathways involved in fibrosis, inflammation, and hypertrophy. Preclinical studies demonstrate that H19 modulates fibrotic pathways by regulating TGF- β signaling and ECM deposition.⁶⁶ As a consequence, pa-tients with higher circulating levels of H19 may experi-ence accelerated fibrosis, contributing to stiffness and impaired contractility of the right ventricle. Thus, H19 serves as a potential prognostic marker and therapeu-tic target for RV dysfunction, with RNA-based inhibitory strategies aimed at attenuating fibrosis and improving RV function.

ECM Protein Biomarkers

ECM proteins are another focus of biomarker research in RV dysfunction, as they are intimately involved in the structural remodeling that characterizes maladaptive hypertrophy and fibrosis in the right ventricle. Two key ECM proteins—SPARCL1 (secreted protein acidic and rich in cysteine-like 1) and CILP1 (cartilage intermedi-ate layer protein 1)—have emerged as novel biomarkers with strong correlations to RV fibrosis and maladaptive remodeling.⁶¹ SPARCL1 is involved in ECM organiza-tion and regulates collagen deposition, which is crucial for maintaining the structural integrity of the right ven-tricle. Elevated levels of SPARCL1 have been linked to increased fibrosis and RV stiffening, impairing systolic and diastolic function. Similarly, CILP1 modulates ECM turnover and fibrosis; increased circulating levels of CILP1 have been observed in patients with PH, particu-larly those with advanced RV failure. Both SPARCL1 and CILP1 hold promise as biomarkers reflecting disease severity and providing insights into the fibrotic damage occurring within the right ventricle myocardium. Given their roles in RV fibrosis, SPARCL1 and CILP1 are being investigated as potential therapeutic targets for antifi-brotic therapies. Additional ECM proteins that have re-cently emerged as potential candidates in RV therapies are galectin-3,⁶⁷ COL18A1 (endostatin),⁶⁸ OPN (osteo-pontin),⁶⁹ and CCN2 (connective tissue growth factor).⁷⁰

Drugs aimed at inhibiting ECM protein synthesis or en-hancing ECM degradation could alleviate the fibrotic bur-den on the right ventricle, thereby preserving its function and improving clinical outcomes in PH.

Inflammatory Biomarkers

Inflammatory biomarkers are also gaining traction as indicators of RV dysfunction, as anticipated in past years.⁷¹ Inflammation, both local in the heart and sys-temic throughout the body, plays a pivotal role in driv-ing RV dysfunction and fibrosis, particularly under the chronic pressure overload conditions associated with PH. Together with TNF- α , IL-6, and IL-1 β , a particularly promising biomarker of inflammation in RV failure is FSTL3 (follistatin-like 3), a glycoprotein that reflects ac-tive inflammation and fibrosis in the myocardium.^{72,73} El-evated levels of FSTL3 have been associated with poor outcomes in patients with PH, particularly those with advanced RV dysfunction. Given its role in mediating fibrosis, FSTL3 has emerged as a notable therapeutic target.⁷² Early clinical trials have explored the efficacy of anti-inflammatory agents that inhibit activin signal-ing, a pathway closely linked to FSTL3 activity. These agents have shown promise in reducing RV fibrosis and improving contractility,¹⁶ thus highlighting the therapeu-tic potential of targeting local inflammation in RV fail-ure. Furthermore, antifibrotic therapies that specifically target inflammatory signaling pathways, such as TGF- β or interleukin pathways, represent a new frontier in RV failure management (Table 1).^{20–24,27,30,31,34–36,42–46,48–50,74} By attenuating the inflammatory response, these thera-pies aim to slow or reverse RV remodeling, thus pre-serving cardiac function and delaying the progression to HF.

FUTURE DIRECTIONS

Improving the Prognostic Value of Advanced Imaging Techniques and Looking Beyond RV

Guidelines from the European Society of Cardiology and European Respiratory Society incorporate imaging met-rics for initial diagnoses, but lack a comprehensive model for ongoing risk assessment in PH. Therefore, large-scale

prospective studies are essential to evaluate the prognostic value of advanced imaging techniques and to incorporate them into simplified risk stratification models (Figure 3). Improved patient management strategies depend on the effective identification of those at risk for progressive disease. A multiparametric approach can optimize the utility of cardiac imaging while addressing the limitations associated with individual measures. Recent research has identified distinct phenotypes by combining various echocardiographic measures, leading to more accurate prognostic predictions for patients with PH.⁷⁵

To better capture the complexity of the right ventricle, future studies should not only have an RV-centric approach but also integrate advanced measures, including strain analysis, 3-dimensional echocardiography, atrial function, LV function, and myocardial work evaluation.⁵⁵ Recent findings suggest that combining novel measures enhances risk detection, although further validation in larger cohorts is warranted to establish their clinical utility. The addition of the LV eccentricity index—a ratio of the major and minor LV diameters in the short axis plane—to the REVEAL Lite 2 risk score calculator, for example, provided more accurate risk predictions than REVEAL Lite 2 alone.⁷⁶ Patients with severe PH have increased prevalence of LV diastolic dysfunction—associated septal displacement, LV compression, and possibly decreased LV compliance. There is a growing

appreciation of non-RV-related factors contributing to right HF in PH, including LV dysfunction,⁷⁷ atrial function,⁷⁸ and right atrial-ventricular coupling.⁷⁹ The emergence of automated or semiautomated segmentation of the heart chambers^{80,81} will facilitate the incorporation of these non-RV-related factors in a robust model for early detection of right HF. By using a comprehensive approach to imaging-based phenotyping, we can deepen our understanding of RV adaptation and maladaptation, ultimately leading to improved management strategies for patients with PH. This integrated approach not only enhances diagnostic precision but also opens new avenues for targeted therapeutic interventions aimed at preserving RV function and improving patient outcomes through informed clinical decision-making.

Multimomics Approaches and Identification of Disease-Specific and RV State-Specific Biomarkers

The identification of novel biomarkers for RV dysfunction and failure in PH is rapidly becoming an active area of research, as it holds promise for improving early diagnosis, disease monitoring, and therapeutic intervention. Biomarkers specific to RV pathology may be used to refine risk stratification, track disease progression, and inform personalized treatment strategies.

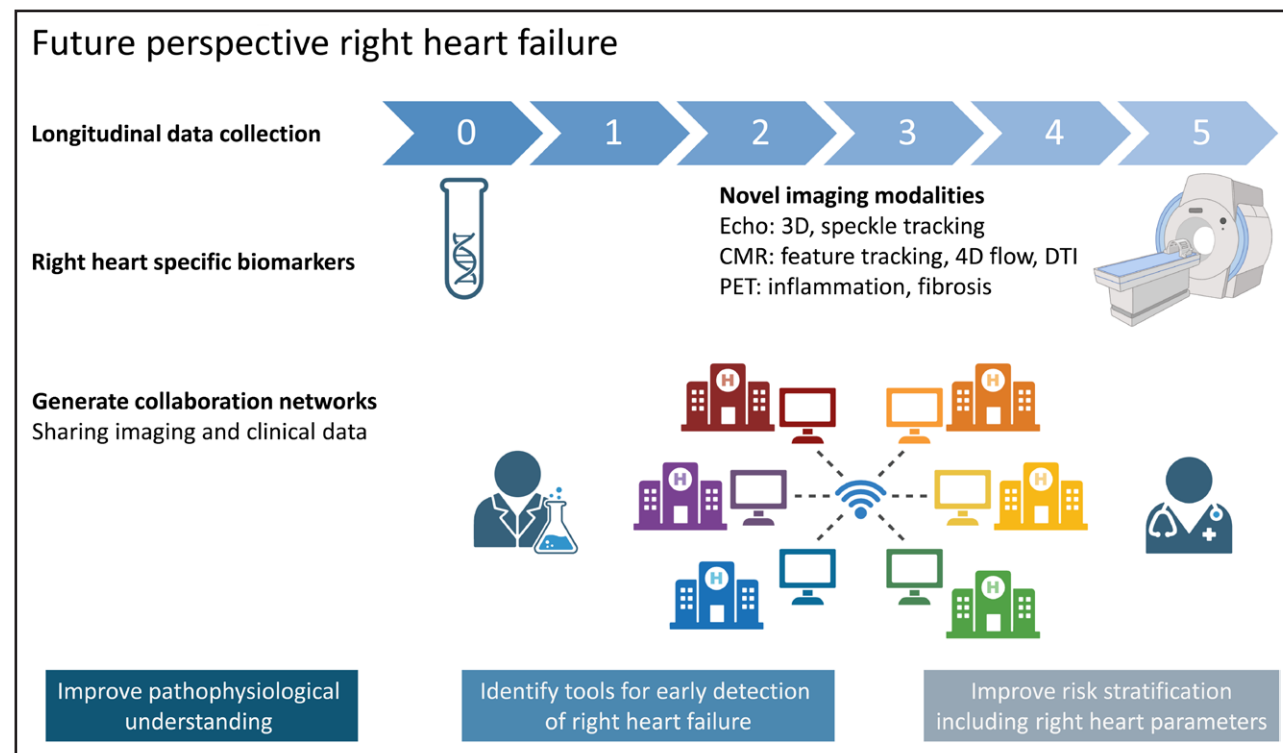


Figure 3. Advancing right heart phenotyping in pulmonary hypertension through longitudinal data and imaging innovation.

Emerging echocardiographic (eg, 3-dimensional [3D] imaging, speckle tracking), cardiac magnetic resonance (CMR; feature tracking, 4-dimensional [4D] flow, diffusion tensor imaging [DTI]), and positron emission tomography (PET)-based techniques (targeting inflammation and fibrosis) provide novel insights into RV structure and function. These approaches aim to improve early detection of RV failure, enhance risk stratification incorporating right heart measures, and foster collaborative networks for imaging and clinical data sharing.

Recent studies integrating transcriptomics, proteomics, and metabolomics have identified distinct molecular signatures linked to adaptive and maladaptive RV remodeling in both animal models and human patients with PH (Tables 2 and 3).^{3,61,82–99} Studies integrating transcriptome analysis of human RV tissues and plasma proteomic profiling of circulating biomarkers in 2 independent PAH cohorts have identified NID1 (nidogen 1), C1QTNF1 (C1q and TNF related 1), and CRTAC1 (cartilage acidic protein 1) as predictive biomarkers of maladaptive states of the right ventricle that correlate with magnetic resonance imaging measures and poor clinical outcomes.⁶¹ By integrating transcriptome and proteome profiling of RV tissues with plasma proteomic analysis, a fibrosis-related protein, LTBP-2 (latent

transforming growth factor β binding protein 2), was identified as an important biomarker that correlates with RV function and improves risk stratification for predicting long-term survival in PAH.⁹⁹

Inflammatory proteomics has also uncovered circulating biomarkers—such as CRIM1 (cysteine-rich transmembrane BMP regulator 1), FSTL3, and SPON1 (spondin-1)—that improve risk stratification in PAH.⁷² Integration of metabolomics and transcriptomics identified lactate as a key metabolic feature of RV failure, detectable in both RV tissue and plasma.⁶³ Comparative multispecies analyses have also shown that porcine RV models closely resemble human RV dysfunction, particularly with respect to changes in metabolic pathways.⁸⁶ As multiomics research advances, these findings will drive

Table 2. Right Ventricular Omics Studies in Animal Models of Pulmonary Hypertension

Models and tissues	Omics	Results	References
4 Sham and 4 PAC mice (3 wk), right and left ventricles	Microarray	Identification of RV remodeling biomarkers, including CILP	81
Sham, PAB, and debanded mice at different time points (n=2), right ventricle	Microarray	Changes in inflammatory and remodeling pathways	82
3 Sham and 3 PAB (1 h) rats, right and left ventricles	Bulk RNA-seq	Changes in inflammatory and remodeling pathways*	83
5 Sham, 5 compensated PAB vs 5 decompensated PAB, right ventricle	Bulk RNA-seq	Changes in pathways related to inflammation and metabolism; identification of biomarkers for RV decompensation	61
28 Rats grouped in sham, compensated PAB (7 wk), and decompensated PAB (14 wk) groups, right and left ventricles	Bulk RNA-seq (whole heart and cardiomyocytes), proteomics	Changes in pathways related to inflammation and ECM	84
4 Sham and 4 PAB pigs (8 wk after PAB), right ventricle	Bulk RNA-seq, proteomics	Alterations in pathways related to fatty acid metabolism and dilated cardiomyopathy*	85
46 Pigs treated with sham, pulmonary vein banding, aortopulmonary shunting, or PAB, plasma	Proteomics and metabolomics	Changes in fatty acid metabolism and high-density lipoprotein components	86
Normoxic and hypoxic rats (2 wk), right and left ventricles	Bulk RNA-seq	Changes in pathways related to inflammation*	87
Normoxic and hypoxic rats (in utero plus 6 wk), right ventricle and atrium	Bulk RNA-seq	Changes in inflammatory and remodeling pathways†	88
25 Normoxic and 29 hypoxic (2 wk) calves, right ventricle	Microarray	Changes in pathways related to inflammation and ECM*	89
15 Control and 8 MCT (4 wk) rats, right ventricle	Bulk RNA-seq	Changes in pathways related to inflammation and oxidative metabolism*	90
7 Control, 7 MCT (4 wk), and 7 HxSu (3 wk + 2 wk normoxia), right ventricle	Bulk RNA-seq	Changes in pathways related to inflammation and oxidative metabolism*	91
Control and MCT rats, fecal samples	Metabolomics on intestinal microbiome	Changes in FAO and glutaminolysis*	92
5 Control and 3 MCT (4 wk) rats, right ventricle	Proteomics	Changes in proteins related to metabolism and cellular respiration*	93
30 Rats grouped in control, compensated MCT, and decompensated MCT groups, right ventricle	Bulk RNA-seq	Changes in pathways related to inflammation and metabolism; identification of biomarkers for RV decompensation	61
Rats injected in jugular vein with 2 doses of beads, and killed at different time points, right ventricle	Microarray	Changes in pathways related to inflammation and oxidative metabolism	94
HxSu rats (3 wk followed by 5 wk normoxia), right ventricle	Bulk RNA-seq, metabolomics	Changes in proteins related to metabolism and cellular respiration*	95

CILP indicates cartilage intermediate layer protein; ECM, extracellular matrix; FAO, fatty acid oxidation; HxSu, hypoxia/SUGEN5416; MCT, monocrotaline; PAB, pulmonary artery banding; PAC, pulmonary artery constriction; RNA-seq, RNA sequencing; and RV, right ventricular.

*Compensation not studied.

†Sex considered.

Table 3. Right Ventricular Omics Studies in Patients With Pulmonary Arterial Hypertension

Patients	Omics	Results	References
88 Controls, 76 patients with PAH	Plasma proteomics	Identification of RV failure biomarkers	96
5 Controls, 5 patients with LV failure, 5 patients with both RV and LV failure	Bulk RNA-seq	Identification of potential determinants of RV failure, including WIPI1*	97
14 Controls, 11 patients with compensated PAH, 7 patients with decompensated PAH	Bulk RNA-seq, proteomics and plasma proteomics	Changes in pathways related to inflammation and metabolism; identification of biomarkers for RV decompensation	98
40 Patients, in control, compensated PAH, and decompensated PAH groups	Bulk RNA-seq, plasma proteomics	Changes in pathways related to inflammation and metabolism; identification of biomarkers for RV decompensation	61
20 Controls, 41 patients with PAH	Plasma metabolomics, nanostring	Increase of glycolytic pathways in patients with PAH	63

LV indicates left ventricular; PAH, pulmonary arterial hypertension; RNA-seq, RNA sequencing; RV, right ventricular; and WIPI1, WD repeat domain, phosphoinositide-interacting 1.
*Compensation not studied.

precision medicine approaches, offering novel therapeutic avenues tailored to the molecular mechanisms underlying RV dysfunction in PH.

Integrating Molecular Biomarkers With Advanced Imaging Techniques

Molecular analyses of tissue samples can provide key insights into the pathways activated during the transition from adaptive to maladaptive remodeling, thereby enhancing our understanding of the underlying pathophysiology and facilitating individual molecular phenotyping. Future large-scale studies should aim to integrate

imaging and molecular phenotyping to provide a more comprehensive understanding of disease pathogenesis and enable personalized treatment strategies. The convergence of multiomics approaches encompassing genomics, transcriptomics, proteomics, and metabolomics with sophisticated imaging modalities has the potential to revolutionize diagnostic precision and therapeutic decision-making.

Linking molecular data with high-resolution cardiac imaging presents a powerful opportunity to improve RV phenotyping. Molecular signatures identified through multiomics, including genetic variants, transcriptomic profiles, or protein expression patterns, can be correlated

Table 4. Imaging Studies for Detection of Right Ventricular Maladaptation in Patients With Pulmonary Hypertension

Imaging techniques and measures	Notes	References
Cardiac MRI		
RVEF	High EF associated with survival	99
SV/ESV	Low SV/ESV associated with worse prognosis	100
SVI	High SVI associated with survival	101
Mass–volume index	High mass–volume index associated with less clinical worsening	102, 103
Mass–volume index	High mass–volume index associated with survival	104
ESVI	ESVI associated with mortality	105
Echocardiography		
TAPSE	TAPSE associated with survival	106
Right atrial area	Right atrial area associated with mortality	107
FAC	FAC associated with survival	108
TAPSE/sPAP	TAPSE/sPAP associated with survival	109
GLS/sPAP	GLS/sPAP associated with survival	110
S′/RA area index	S′/RA area index as a marker of diastolic dysfunction associated with mortality	111
SV/ESV	SV/ESV with calculation of effective SV associated with survival	112
TAPSE/sPAP, RV end diastolic area, LV end diastolic area	Different phenotypes associated with mortality	74

EF indicates ejection fraction; ESV, end-systolic volume; ESVI, end-systolic volume index; FAC, right ventricular fractional area change; GLS, global longitudinal strain; MRI, magnetic resonance imaging; RA, right atrial; RVEF, right ventricular ejection fraction; S′, lateral tricuspid annulus peak systolic velocity; sPAP, systolic pulmonary artery pressure; SV, stroke volume; SVI, stroke volume index; and TAPSE, tricuspid annular plane systolic excursion.

with advanced imaging features, such as RV free wall strain, 3D echocardiography, and CMR (Table 4).^{55,75,100–113} By combining these modalities, clinicians will be better equipped to phenotype the right ventricle accurately, distinguish between adaptive and maladaptive remodeling, and identify patients who may benefit from targeted interventions. The recent discovery of *BMPR1A* genetic variants that predict better RV function, as assessed by CMR, in women with PAH exemplifies this approach by combining large omics data sets (>30 000 patients) with imaging.⁴⁷

Emerging technologies, such as artificial intelligence (AI) and machine learning, are poised to play a key role in the future of RV dysfunction management. These tools can analyze large data sets from molecular and imaging studies, generating predictive models that identify patients at risk for RV failure. AI-driven algorithms could help stratify patients on the basis of their molecular profiles, guiding personalized treatment plans that target the specific pathways driving their RV dysfunction. The integration of molecular and imaging data also holds promise for clinical trials, enabling better-tailored patient selection and therapeutic responses based on individual disease mechanisms. This personalized approach signifies a shift in the management of RV dysfunction and may ultimately improve outcomes for patients with PH.

Current studies are largely descriptive, identifying correlations between molecular features and phenotypes. Moving forward, a more integrative systems biology approach will be essential. By applying network-based analyses, pathway enrichment, and machine learning, researchers can develop causal inference models that move beyond associations to uncover regulatory pathways driving RV remodeling. These methods enable the identification of upstream effectors and network hubs offering deeper mechanistic insights and novel therapeutic targets.

Emerging Role of AI and Machine Learning in RV Dysfunction

With the growing complexity of omics and imaging data, AI and machine learning are poised to play a transformative role in RV dysfunction research and care. These tools can integrate diverse datasets to build predictive models, identify molecular-imaging correlates, and stratify patients based on their individual risk and pathobiology. AI can help uncover hidden patterns and causal relationships within high-dimensional datasets facilitating the generation of testable hypotheses and more targeted therapeutic strategies.

The integration of AI with systems biology and multimodal data will also optimize patient selection and outcome prediction in clinical trials, accelerating the development of precision therapies for RV failure in

PH. Ultimately, these innovations represent a paradigm shift toward mechanistically driven, personalized care for patients with RV dysfunction.

Developing Personalized Treatment Approaches for RV Dysfunction and Maladaptive Remodeling

The development of personalized treatment strategies for RV dysfunction and maladaptive remodeling hinges on experimental systems that replicate the complexity of human RV pathology. In recent years, several innovative platforms have emerged, offering unprecedented opportunities to investigate RV-specific disease mechanisms and to test targeted therapies in patient-derived models.

Induced pluripotent stem cells, RV organoids, and living RV slices are 3 such tools that enable functional, genetic, and metabolic exploration of RV remodeling. Induced pluripotent stem cells, reprogrammed from somatic cells from people with PH, can be differentiated into cardiomyocytes, endothelial cells, and fibroblasts. This facilitates modeling of genetic susceptibilities and molecular drivers of RV remodeling, enabling high-throughput drug screening, gene-editing interventions (eg, CRISPR-Cas9), and identification of patient-specific therapeutic targets. RV organoids—3-dimensional multicellular constructs—replicate the right ventricle microenvironment and cellular heterogeneity. These platforms allow assessment of cell–cell interactions, ECM remodeling, and responses to mechanical stress or hypoxia while serving as tools to investigate fibrosis, inflammation, and metabolic reprogramming under controlled conditions.¹¹⁴ Living RV slices (ie, functional ex vivo myocardial tissue) retain native architecture, electrophysiology, and contractile function. These slices are ideal for real-time analysis of fibrosis, metabolic shifts, and immune cell infiltration in response to pharmacologic agents or genetic modulation.¹¹⁵

Beyond these platforms, emerging bioengineering and computational models offer new frontiers. A recent study by Singh et al¹¹⁶ introduced a robotic right ventricle platform, which integrates biomechanical stimulation with human cardiac tissue to precisely simulate RV pressure overload and test mechanical and pharmacologic interventions. This system enables real-time assessment of contractile reserve, tissue stiffness, and mechanotransduction pathways under programmable loading conditions. Such tools bridge the gap between in vitro models and in vivo physiology, offering a powerful translational bridge for therapy development.

By integrating these experimental platforms with multiomics technologies (eg, genomics, transcriptomics, proteomics, metabolomics) and advanced imaging modalities (eg, 3D echocardiography, CMR, PET), researchers can develop comprehensive, individualized

RV disease models. These models facilitate targeting of patient-specific mechanisms, such as mitochondrial dysfunction, metabolic shifts, or maladaptive fibrosis, through precision therapies, including RNA-based interventions, epigenetic modifiers, or small molecule inhibitors.

Many of these platforms are still maturing, but the field of right heart research can accelerate progress by leveraging well-established methodologies from left heart studies. Translational insights from LV disease modeling, drug development, and biomarker discovery will be essential to refine right heart-specific strategies and drive clinical implementation.

CONCLUSION

RV dysfunction is a key determinant of outcomes in PH. The complex interplay of hemodynamic stress, inflammation, metabolic reprogramming, and molecular signaling drives the progression from adaptive remodeling of the right ventricle to eventual failure. Recognizing the mechanisms that underlie RV dysfunction is essential not only for informing clinical practice but also for identifying novel therapeutic targets that may improve patient outcomes. Ongoing research into biomarkers, molecular pathways, and advanced imaging techniques holds promise for the development of more effective treatments aimed at preserving RV function and preventing HF. By integrating these approaches, clinicians can better tailor interventions, leading to enhanced management strategies. Furthermore, a deeper understanding of these multifaceted interactions may unlock new avenues for targeted therapies that

address the root causes of RV dysfunction, ultimately driving improvements in prognosis and quality of life for patients with PH. As we continue to advance our understanding of RV pathophysiology, the potential for innovative treatments and more personalized patient care approaches becomes increasingly within reach.

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The American Heart Association makes every effort to avoid any actual or potential conflicts of interest that may arise as a result of an outside relationship or a personal, professional, or business interest of a member of the writing panel. Specifically, all members of the writing group are required to complete and submit a Disclosure Questionnaire showing all such relationships that might be perceived as real or potential conflicts of interest.

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Writing group member	Employment	Research grant	Other research support	Speakers' bureau/honoraria	Expert witness	Ownership interest	Consultant/advisory board	Other
Soni Savai Pullamsetti	Universities of Giessen and Marburg Lung Center (UGMLC), Institute for Lung Health (ILH), Cardio-Pulmonary Institute (CPI), Member of the German Center for Lung Research (DZL), Justus Liebig University (Germany) Max Planck Institute for Heart and Lung Research, Bad Nauheim (Germany)	German Research Council (DFG) CRC1213 A01, A05*	None	None	None	None	None	None

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Writing Group Disclosures Continued

Writing group member	Employment	Research grant	Other research support	Speakers' bureau/honoraria	Expert witness	Ownership interest	Consultant/advisory board	Other
Sébastien Bonnet	Université Laval Quebec Heart and Lung Institute (Canada) Universities of Giessen and Marburg Lung Center (UGMLC), Institute for Lung Health (ILH), Cardio-Pulmonary Institute (CPI), Member of the German Center for Lung Research (DZL), Justus Liebig University (Germany)	CIHR, FRQS and IUCPQ foundation*	None	None	None	None	None	None
Vinicio A. de Jesus Perez	Stanford University	R01HL139664, R01HL134776, R01HL59886, R01HL160018, 5R01HL172449-02, and Dean's Endowment Funds*	None	None	None	None	None	None
Frances S. de Man	VU University Medical Center/Institute of Cardiovascular Research Pulmonology (Netherlands)	None	None	None	None	None	None	None
Anna R. Hemnes	Vanderbilt University	NHLBI 1K24HL155891-01A1*; NHLBI UG3 HL175041 – 01*; NHLBI 2 R01 HL 142720-01A1*	None	None	None	Tenax Therapeutics*	Gossamer-Biot; Merck; Johnson & Johnson*; United Therapeutics	None
Laura Mercer-Rosa	Children's Hospital of Philadelphia	None	None	None	None	None	None	None
Monica Mukherjee	Johns Hopkins University School of Medicine	NIH/NHLBI R01HL162851†; US Department of Defense PR231648†	None	None	None	None	AllRock Bio, Inc.*; Advarra, Inc.*	None
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Khodr Tello	Justus Liebig University; Universities of Giessen and Marburg Lung Center (UGMLC); member of the German Center for Lung Research (Germany)	German Research Council (DFG) CRC1213 B8*	None	None	None	None	None	None
Rebecca R. Vanderpool	The Ohio State University	None	None	None	None	None	None	None

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*Modest.

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Reviewer	Employment	Research grant	Other research support	Speakers' bureau/honoraria	Expert witness	Ownership interest	Consultant/advisory board	Other
Paul Hassoun	Johns Hopkins University	NIH/NHLBI (R01HL114910: Mechanisms of RV dysfunction in scleroderma-associated PH)†	None	None	None	None	MSD*	None
Tim Lahm	National Jewish Health/ University of Colorado	None	None	None	None	None	None	None
Jane A. Leopold	Brigham and Women's Hospital	None	None	<\$5000*	None	None	None	None
Kurt Prins	Medical University of South Carolina	None	None	None	None	None	None	None
Gopinath Sutendra	University of Alberta (Canada)	None	None	None	None	None	None	None

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*Modest.
†Significant.

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